

18.06 Linear Algebra - Lectures 23 & 24
 Textbook Problem - Sections 6.3 & 6.4

(1-9)[3] (10-13)[2] (18-27)[2]

Section 6.3 - Problem Set

1) $\frac{du}{dt} = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix} u$ $u = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2$
 $u(0) = (5, -2)$

$A = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix}$ $C = X^{-1} \cdot u(0)$

$A = X \Lambda X^{-1}$ $\det(A - \lambda I) = 0$
 $(4 - \lambda)(1 - \lambda) + (0)(3) = 0$
 $\lambda = 4 \text{ or } 1$

$(A - \lambda I)x = 0$
 $\begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} x = 0$ $x_1 = 1$ $x_2 = 0$ $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$;

$\lambda = 4$
 $\begin{bmatrix} 0 & 3 \\ 0 & -3 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} x = 0$ $x_{s1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda = 1$
 $\begin{bmatrix} 3 & 3 \\ 0 & 0 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} x = 0$ $x_{s2} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
 $x_1 + (1) = 0$

$X = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ $X^{-1} = \frac{1}{-(1)(1) - 0} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$

$C = X^{-1} u(0) = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \end{bmatrix}$
 $= \begin{bmatrix} -5 + (-2) \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ -2 \end{bmatrix}$

$u(0) = \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
 $u = 3e^{4t} x_1 + -2e^{1t} x_2$
 $= 3e^{4t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2e^t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

4) $V(0) = 30$ people
 $w(0) = 10$ people

Movement: $V \rightarrow w$

$$\frac{dv}{dt} = w - v$$

$$\frac{dw}{dt} = v - w$$

$$\frac{du}{dt} = Au$$

$$\rightarrow u = (v, w) \quad \text{so} \quad A = \begin{bmatrix} -1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(-1 - \lambda)(-1 - \lambda) - (1)(1) = 0$$

$$1 + 2\lambda + \lambda^2 - 1 = 0 \quad \lambda^2 + 2\lambda + 0 = 0$$

$$\lambda^2 + 2\lambda + 2 = 0$$

$$\lambda(\lambda + 2) = 0$$

$$\lambda \in \{$$

$$\lambda = 0 \text{ or } -2$$

$$\lambda = 0:$$

$$\lambda = -2:$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0$$

$$\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

$$\text{set } x_2 = 1$$

$$-x_1 + (1) = 0$$

$$x_1 = 1$$

$$x_s = \begin{bmatrix} +1 \\ +1 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0$$

$$\text{set } x_2 = 1$$

$$x_1 + (1) = 0$$

$$x_1 = -1$$

$$x_{s2} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad X^{-1} = \frac{1}{(1)(1) - (-1)(1)} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

$$C = X^{-1} u(0)$$

$$= \begin{bmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 30 \\ 10 \end{bmatrix} = \begin{bmatrix} 20 \\ -10 \end{bmatrix}$$

$$u(t) = (20)e^{(0)t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-10)e^{(-2)t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= 20 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 10e^{-2t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= 20 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 10e^{-2t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

4 con.) total $v+w$:

$$\begin{aligned}\frac{d(v+w)}{dt} &= \frac{dv}{dt} + \frac{dw}{dt} \\ &= (w-v) + (v-w) \\ &= \underline{0} \quad \text{so constant.}\end{aligned}$$

$$u(t) = \begin{bmatrix} 20 \\ 20 \end{bmatrix} + \underbrace{10e^{-2t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\text{decays to zero}} \quad t \rightarrow \infty$$

so total = $20 + 20 = \underline{40 \text{ people}}$ ✓

~~$u(t) = \begin{bmatrix} 20 + 10e^{-2t} \\ 20 - 10e^{-2t} \end{bmatrix}$~~

$v(1) = 20 + 10e^{-2}$

$w(1) = 20 - 10e^{-2}$

$v(\infty) = 20$

$w(\infty) = 20$

8) $\frac{dr}{dt} = 6r - 2w$ $\frac{dw}{dt} = 2r + w$ $r(0) = w(0) = 30$

$u = (r, w)$
 ~~$\frac{du}{dt} = Au$~~ $A = \begin{bmatrix} 6 & 2 \\ -2 & 1 \end{bmatrix}$

$\det(A - \lambda I) = 0$:

$(6-\lambda)(1-\lambda) - (2)(-2) = 0$

$6 - 7\lambda + \lambda^2 + 4 = 0$

$\lambda^2 - 7\lambda + 10 = 0$

$(\lambda - 2)(\lambda - 5) = 0$

$\lambda = 2 \text{ or } 5$

$\lambda = 2$:

$\begin{bmatrix} 4 & 2 \\ -2 & -1 \end{bmatrix} x = 0$
 $\begin{bmatrix} 4 & 2 \\ 0 & 0 \end{bmatrix} x = 0$

set $x_2 = 1$, $4x_1 + 2(1) = 0$

$x_{s1} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$x_1 = \frac{-2}{4} = -\frac{1}{2}$

$\lambda = 5$

$\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} x = 0$

$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} x = 0$

set $x_2 = 1$, $x_1 + 2(1) = 0$

$x_1 = -2$

$x_{s2} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow X = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix}$

$X^{-1} = \frac{1}{(-1)(1) - (-2)(2)} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$= \frac{1}{-1 - (-4)} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$= \frac{1}{3} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix}$

$$X^{-1} = \begin{bmatrix} 1/3 & 2/3 \\ -2/3 & -1/3 \end{bmatrix} \quad \times$$

$$C = X^{-1} u(0)$$

$$= \begin{bmatrix} 1/3 & 2/3 \\ -2/3 & -1/3 \end{bmatrix} \begin{bmatrix} 30 \\ 30 \end{bmatrix}$$

$$= \begin{bmatrix} 30 \\ -30 \end{bmatrix} \} \times \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 30 \\ 30 \end{bmatrix} \quad \text{e}_1$$

$$u(t) = (30)e^{5t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + (-30)e^{2t} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$= 30e^{5t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 30e^{2t} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$A x_1 = \lambda_1 x_1$$

$$\begin{bmatrix} 6 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} \quad ; \quad \begin{bmatrix} 6 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$$

2x2 2x1

$$\lambda_1 x_1 = (2) \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} ; \quad \lambda_2 x_2 = (5) \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \end{bmatrix}$$

$$X^{-1} = \frac{1}{(2)(2) - (1)(1)} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

$$C = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 30 \\ 30 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

$$u(t) = 10e^{5t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 10e^{2t} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$u(t) = 10e^{5t} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 10e^{2t} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$r(t) = 20e^{5t} - 10e^{2t}$$

$$w(t) = -10e^{5t} + 20e^{2t}$$

* Eigenvectors were wrong
Swapped signs!

$$10) \quad y'' = 5y' + 4y \quad u = (y, y')$$

$$\frac{du}{dt} = \begin{bmatrix} y' \\ y'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} y \\ y' \end{bmatrix} = Au$$

$$A = \begin{bmatrix} 0 & 1 \\ 4 & -5 \end{bmatrix}$$

$$\det(A - I\lambda) = 0$$

$$(0 - \lambda)(-5 - \lambda) - (1)(4) = 0$$

$$-5\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 5\lambda - 4 = 0$$

~~$$\lambda = \frac{5 \pm \sqrt{41}}{2}$$~~

$$\lambda = \frac{1}{2}(5 \pm \sqrt{41})$$

} ✓

sub in $y = e^{\lambda t}$

$$y'' = 5y' + 4y$$

$$(e^{\lambda t})'' = 5(e^{\lambda t})' + 4e^{\lambda t}$$

$$\lambda^2 e^{\lambda t} = 5\lambda e^{\lambda t} + 4e^{\lambda t}$$

$$e^{\lambda t}(\lambda^2 - 5\lambda - 4) = 0$$

$$\lambda = \frac{1}{2}(5 \pm \sqrt{41})$$

} ✓

$$19) B = \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} \quad B^2 = 0$$

$$e^{Bt} = e^{\begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} t}$$

$$= I + Bt + \frac{1}{2}(Bt)^2 + \frac{1}{6}(Bt)^3$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} t + \frac{1}{2} \left(\begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} t \right)^2 + \frac{1}{6} \left(\begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} t \right)^3$$

$$= I + Bt$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} t$$

$$= \begin{bmatrix} 1 & -4t \\ 0 & 1 \end{bmatrix}$$

$$(e^{Bt})' = \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} = B$$

$$= B e^{Bt}$$

$$= \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -4t \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix}$$

$$= B$$

$$= \begin{bmatrix} 0 & -4 \\ 0 & 0 \end{bmatrix} e^{Bt}$$

$$21) A = \begin{bmatrix} 1 & 4 \\ 0 & 0 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(1 - \lambda)(0 - \lambda) - (4 \times 0) = 0$$

$$0 - 0 + \lambda^2 - \lambda - 0 = 0$$

$$\lambda(\lambda - 1) = 0$$

$$\lambda = 0 \text{ or } 1$$

$$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(A - \lambda I)x = 0$$

$$\lambda = 0$$

$$\begin{bmatrix} 1 & 4 \\ 0 & 0 \end{bmatrix} x = 0$$

$$\text{Set } x_2 = 1$$

$$x_1 + 4(1) = 0$$

$$x_1 = -4$$

$$\begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{(1)(0) - (0)} \begin{bmatrix} -1 & -4 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$$

$$\lambda = 1$$

$$\begin{bmatrix} 0 & 4 \\ 0 & -1 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} x = 0$$

$$\text{Set } x_1 = 1$$

$$4x_2 = 0$$

$$x_2 = 0$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = X \Lambda X^{-1}$$

$$= \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$$

$$e^{At} = X e^{\Lambda t} X^{-1}$$

$$= \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$$

Problem Set 6.4

1) $B = -A =$
 B is an antisymmetric matrix

$$\begin{array}{lll} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} & A \Rightarrow A^T & \text{symmetric} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} & A \Rightarrow A^{-1} & \text{orthogonal not symmetric} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} & A \Rightarrow A^T, A^{-1} \rightarrow A^T & A^T \rightarrow A^{-1} \end{array}$$

③ Symmetric with same eigenvalues as S .
 $B = A^T = A^{-1}$ to be orthogonal ✓

5) $S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\det(S - \lambda I) = 0$

~~$S - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{bmatrix}$~~

~~$\det(S - \lambda I) = (2)(-1)^{1+3} \cdot ((0)(0) - (2)(-\lambda)) + (-\lambda)(-1)^{3+3}((-1)(-\lambda) - (0)(0))$~~

~~$= -2(2)\lambda = -4\lambda = 0$~~

~~$= 4\lambda + (-\lambda)(\lambda^2)$~~

~~$= 4\lambda - \lambda^3$~~

~~$= \lambda(4 - \lambda^2) = \lambda(2 - \lambda)(2 + \lambda)$~~

~~$= (-\lambda)(-1)^{1+1} [(-\lambda)^2 - (0)] + (2)(-1)^{1+3} [(0) - (-\lambda)(2)]$~~

~~$= -\lambda^3 + 2(2\lambda)$~~

~~$= -\lambda^3 + 4\lambda$~~

~~$= \lambda(4 - \lambda^2) = \lambda(2 - \lambda)(2 + \lambda)$~~

$$6) S = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$$

$$S^T = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$$

$$S = S^T \quad \text{O}$$

$$\det(S - \lambda I) = 0$$

$$(-2 - \lambda)(7 - \lambda) - (6)^2 = 0$$

$$-14 + 2\lambda - 7\lambda + \lambda^2 - 36 = 0$$

$$\lambda^2 - 5\lambda - 50 = 0$$

$$(\lambda + 5)(\lambda - 10) = 0$$

$$\lambda = -5 \text{ or } 10$$

Φ

$$\lambda = -5:$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} x = 0$$

$$\begin{bmatrix} 3 & 6 \\ 0 & 0 \end{bmatrix} x = 0$$

$$\text{set } x_2 = 1$$

$$3x_1 + 6(1) = 0$$

$$x_1 = -2$$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\lambda = 10$$

$$\begin{bmatrix} -12 & 6 \\ 6 & -3 \end{bmatrix} x = 0$$

$$\begin{bmatrix} -12 & 6 \\ 0 & 0 \end{bmatrix} x = 0$$

$$\text{set } x_2 = 1$$

$$-12x_1 + 6(1) = 0$$

$$x_1 = +\frac{1}{2}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$Q = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{X Normalize to Unit Vectors}$$

$$F = \sqrt{(2)^2 + (1)^2}$$

$$= \sqrt{5}$$

$$Q = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$10) A^3 = 0$$

$$A = X \Lambda X^{-1} \quad \Lambda^3 = 0$$

$$A^k = X \Lambda^k X^{-1}$$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} \lambda^2 & 0 \\ 0 & \lambda^2 \end{bmatrix}$$

$$\lambda^3 = 0$$

$$\lambda = 0$$

~~A is singular~~

All λ s are $= 0$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

If A symmetric :

~~A~~

$$A^3 = Q \Lambda^3 Q^T = 0$$

$$\Lambda = 0$$

only symmetric A is $Q \mathbf{0} Q^T = \text{zero matrix}$

$$16) \text{ Antisymmetric } M^T = -M$$

$$|\lambda| = 1 \quad \therefore M \text{ also } \underline{\text{orthogonal}}. \quad \checkmark$$

eigenvalues either $\pm i$:

$$\text{trace} = 0 + 0 + 0 + 0 = 0$$

$$\sum \lambda_i = 0$$

$$\lambda = \{ +i, -i, +i, -i \} \quad \checkmark$$

$$21) S = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det(S - \lambda I) = 0 \quad \det S = (1)(-1)^{(2+1)}((1)(1) - (0)(0))$$

$$= (-1)(1) = -1$$

character

$$\begin{bmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix} \quad \det(S - \lambda I) = (-\lambda)(-1)^{(1+1)}[(-\lambda)(1-\lambda)] +$$

$$(1)(-1)^{(1+2)}[(1)(1-\lambda) - 0]$$

$$= -\lambda[-\lambda + \lambda^2] + [-1 + \lambda]$$

$$= \lambda^2 - \lambda^3 - 1 + \lambda$$

$$= -\lambda^3 + \lambda^2 + \lambda - 1$$

$$\lambda_1 = -1 \text{ or } 1 \text{ or } 1.$$

$$\lambda = 1: \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

Set $x_3 = 1$

$$-x_1 + x_2 = 0$$

$$x_1 = x_2$$

normalize: $\frac{1}{\sqrt{(1)^2 + (1)^2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \leftarrow E_1 = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} : x_1, x_2 \in \mathbb{R} \right\}$

$\Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} x = 0$

$$\lambda = -1: \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} x = 0$$

$$\text{Set } x_2 = 1$$

$$x_1 + (1) = 0 \quad x_1 = -1$$

$$x_3 = 0$$

normalize $\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \checkmark$$

21 con.) $B = \begin{bmatrix} -d & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & d \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\det(B - \lambda I) = \begin{vmatrix} -1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = \text{---}$$

$$= (1-\lambda)(-1)^{(2+2)} [(-1-\lambda)(1-\lambda) - (1)(0)]$$

$$= (-1)(1-\lambda)(1+\lambda)(1-\lambda)$$

$$= \text{---} (-1-\lambda)(1-\lambda)^2 = 0$$

$$\therefore \lambda = 1, 1 \text{ or } -1.$$

$\lambda = -1$:

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} x = 0 \Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow x = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$x_1, x_3 = 0$ set $x_2 = 1$

$\lambda = +1$:

$$\begin{bmatrix} -2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} x = 0$$

set $x_3 = 1$

$$-2x_1 + (1) = 0$$

$$x_1 = 1/2$$

$$\text{---} x_2 = 0$$

$$\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\text{---}$$

$$\text{set } x_2 = 1, x_3 = 0 \Rightarrow x = \begin{bmatrix} -2x_1 + x_3 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 + x_3 = 0$$

$$x_3 = 2x_1$$

x_2 free

$$x = \begin{bmatrix} t \\ s \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} ; \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; v_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}; v_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

v_1, v_2, v_3 are not orthogonal. $v_1 \perp v_3$ but $v_2 \perp v_3$ is not true.

$$26) A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$$

$$A = Q \Lambda Q^T$$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & b \\ 1 & 0-\lambda \end{vmatrix} = 0$$

$$= (2-\lambda)(-\lambda) - b = 0$$

$$-2\lambda + \lambda^2 - b = 0$$

$$\lambda^2 - 2\lambda - b = 0$$

$$(\lambda - a_1)(\lambda + a_2) = 0$$

$$a_2 - a_1 = -2$$

$$(-a_1)(a_2) = (-b)$$

$$a_1 a_2 = b$$

$$-3 + 1$$

$$b = -3$$

$$\lambda^2 - 2\lambda - 3 = 0$$

X

→ $Q \Lambda Q^T$ if $b = 1$

→ get $A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$ symmetric and it falls out
 $A = A^T$ if $b = 1$.

Impossible to diagonalize A?

repeated λ 's → which either
 1. have a complete set of eigenvectors (could get X)
 2. does not have set
 → cannot get X.

e.g. $b = -1$

$\begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$ → repeated λ s
 → incomplete set.

b for A to be Singular:

$$A = \begin{bmatrix} 2 & b \\ 1 & 0 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2(0) - b(1)} \begin{bmatrix} a & -b \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{-b} \begin{bmatrix} 0 & -b \\ -1 & 2 \end{bmatrix}$$

∴ No inverse defined ∴ Singular IF $b = 0$